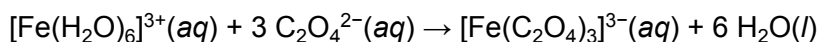


Oxalate anions, $\text{C}_2\text{O}_4^{2-}$, added to a solution of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ displace water molecules within the complex ion to form $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$, as shown in the reaction below:



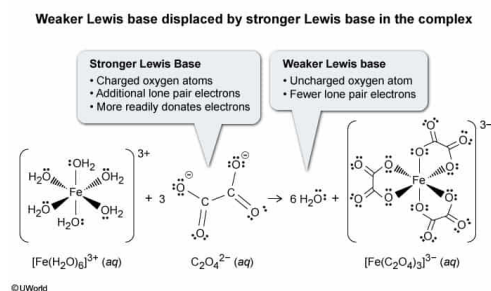
Which of the following does NOT explain why $\text{C}_2\text{O}_4^{2-}$ displaces H_2O in the formation of $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$?

- ✔ ☒ A. $\text{C}_2\text{O}_4^{2-}$ has fewer available lone pairs of electrons per oxygen atom than H_2O . [75%]
- ☐ B. $\text{C}_2\text{O}_4^{2-}$ is a stronger Lewis base than H_2O . [9%]
- ☐ C. $\text{C}_2\text{O}_4^{2-}$ forms coordinate bonds more readily than does H_2O . [7%]
- ☐ D. $\text{C}_2\text{O}_4^{2-}$ is negatively charged whereas H_2O is uncharged. [8%]

Correct

74% answered correctly

Explanation:



Coordination complexes consist of a central metal ion surrounded by one or more ions or molecules called **ligands** that are bound to the metal center by **coordinate bonds**. The ligands act as a **Lewis base**, and the metal center acts as a **Lewis acid**.

Stronger **Lewis bases** can displace weaker Lewis bases as ligands within a complex. Stronger Lewis bases tend to be those that have **lone pair electrons** on atoms with a net charge and/or a lower **electronegativity**. Charged species tend to be less stable (more reactive) than uncharged species because forming a bond produces a more stable, lower energy state. Likewise, atoms with a lower **electronegativity** tend to be better Lewis bases because the electrons are held more loosely and any net charge is less stabilized.

Both $\text{C}_2\text{O}_4^{2-}$ and H_2O have available lone pair electrons on an oxygen atom (same electronegativity aspect), but $\text{C}_2\text{O}_4^{2-}$ is a stronger Lewis base because it also has a negative charge on each of two of the oxygen atoms. This charge comes from the presence of an extra lone pair of electrons on each. Therefore, $\text{C}_2\text{O}_4^{2-}$ does not have *fewer* lone pair electrons per oxygen atom than H_2O , and this choice

does NOT explain why $\text{C}_2\text{O}_4^{2-}$ displaces H_2O .

(Choice B) This statement explains the formation of $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ because weaker Lewis bases tend to form weaker coordinate bonds. $\text{C}_2\text{O}_4^{2-}$ is a stronger Lewis base, which can displace H_2O and form a stronger coordinate bond than H_2O .

(Choices C and D) These choices explain the reaction because charged species tend to be stronger

Lewis bases and more reactive than uncharged species. The two additional lone pairs of electrons on two oxygen atoms in $\text{C}_2\text{O}_4^{2-}$ give $\text{C}_2\text{O}_4^{2-}$ its net charge and a greater tendency to donate electrons for coordinate bond formation.

Educational objective:

Stronger Lewis bases (electron pair donors) can displace weaker Lewis bases as ligands within a coordination complex. Lone pair electrons on atoms with a lower electronegativity tend be stronger Lewis bases than those with a higher electronegativity, and charged atoms with lone pair electrons tend to be stronger Lewis bases than comparable uncharged atoms.

Time spent:0 sec

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